

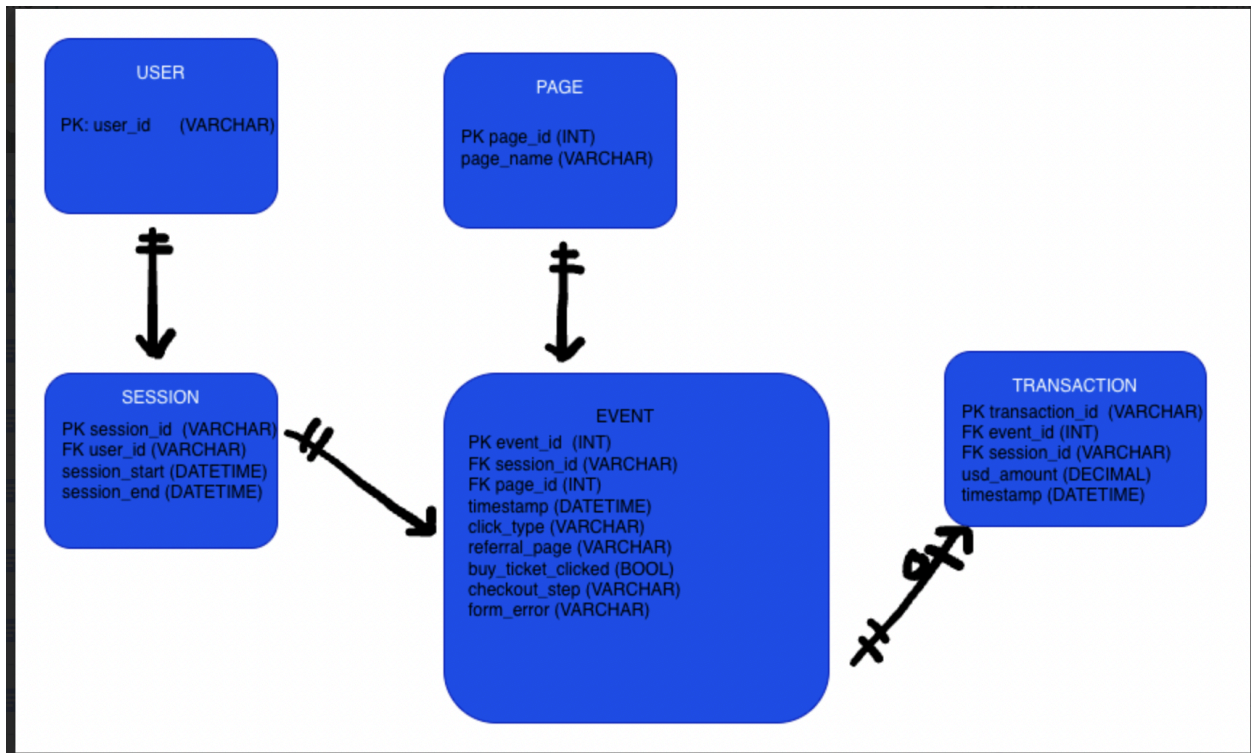
67-250 Technology Project: Phase A

Nandini Tiwari

nandinit

3/11/2026

Database Design Documentation



[Link to ERD in Draw.io](#)

	USER			
	Field	Data Type	Description	Example
PK	user_id	VARCHAR (10)	The user's ID	U0001
	user_type	VARCHAR (50)	The type of user	guest

	SESSION			
	Field	Data Type	Description	Example
PK	session_id	VARCHAR(10)	Unique identifier for a single browsing session	S10001
FK	user_id	VARCHAR(10)	The user <u>that initiated</u> the session	U0572

	TRANSACTION			
	Field	Data Type	Description	Example
PK	transaction_id	INT	Unique identifier for each purchase	2
	timestamp	TIMESTAMP	Date and time of transaction	2025-03-01 11:04:17
	total_amount	DECIMAL(10,2)	Dollar amount charged for purchase	25.00
	ticket_quantity	INT	# tickets purchased in this session	1
FK	session_id	VARCHAR(10)	Reference session in which the purchase occurred	S10008

	PAGEVISIT			
	Field	Data Type	Description	Example
PK	pagevisit_id	INT	Unique identifier for	1

			each page interaction event	
	timestamp	TIMESTAMP	Date and <u>tiem</u> of <u>interaction</u> event	2025-03-01 10:24:01
	click_type	VARCHAR(20)	Type of interaction	page_view
	buy_ticket_clicked	BOOLEAN	Whether user clicked <u>buy</u> ticket button	false
	checkout_step	VARCHAR(20)	Stage of checkout funnel	step_1
	form_error	BOOLEAN	Whether <u>form</u> error happened during this event	false
	usd_amount	DECIMAL(10,2)	Dollar amount of completed purchase (NULL if no transaction happened)	25.00
FK	session_id	VARCHAR(10)	References <u>session</u> this event was in	S10001
FK	page_id	INT	References page being visited	3
FK	referral_page_id	INT	References page user navigated from (NULL if it was the entry page)	1

...

≡

+

PAGE				
	Field	Data Type	Description	Example
PK	page_id	INT	Unique identifier for each page	1
	name	VARCHAR(50)	Internal name of the page	exhibition_detail
	description	VARCHAR(255)	Description of the page's purpose	Detailed view of a museum exhibition

USER

Server: localhost » Database: nandinit » Table: USER

Browse

Structure

SQL

Search

Insert

Export

Import

Operations

Tracking

Triggers

Showing rows 0 - 24 (658 total, Query took 0.0004 seconds.)

SELECT * FROM `USER`

1 > >>

Number of rows: 25

Filter rows: Search this table

Sort by key: None

+ Options

				user_id	user_type
☐	Edit	Copy	Delete	U0001	guest
☐	Edit	Copy	Delete	U0002	student
☐	Edit	Copy	Delete	U0004	member
☐	Edit	Copy	Delete	U0006	guest
☐	Edit	Copy	Delete	U0007	guest
☐	Edit	Copy	Delete	U0008	guest
☐	Edit	Copy	Delete	U0009	guest
☐	Edit	Copy	Delete	U0010	guest
☐	Edit	Copy	Delete	U0011	member
☐	Edit	Copy	Delete	U0012	student
☐	Edit	Copy	Delete	U0013	guest
☐	Edit	Copy	Delete	U0014	student
☐	Edit	Copy	Delete	U0015	guest
☐	Edit	Copy	Delete	U0016	student
☐	Edit	Copy	Delete	U0017	guest
☐	Edit	Copy	Delete	U0018	guest
☐	Edit	Copy	Delete	U0020	guest
☐	Edit	Copy	Delete	U0021	member
☐	Edit	Copy	Delete	U0022	guest
☐	Edit	Copy	Delete	U0023	guest
☐	Edit	Copy	Delete	U0025	guest
☐	Edit	Copy	Delete	U0026	member
☐	Edit	Copy	Delete	U0028	guest
☐	Edit	Copy	Delete	U0030	guest
☐	Edit	Copy	Delete	U0031	member

Server: localhost - Database: nandinit - Table: USER

Browse

Structure

SQL

Search

Insert

Export

Import

Operations

Tracking

Triggers

Table structure

Relation view

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
☐	1	user_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary Unique Index Spatial Fulltext Distinct values Add to central columns
☐	2	user_type	varchar(50)	utf8mb4_bin	No	None		Change Drop Primary Unique Index Spatial Fulltext Distinct values Add to central columns

☐ Check all

With selected:

Browse Change Drop Primary Unique Index Add to central columns Remove from central columns

Print view

Propose table structure

Track table

Move columns

Improve table structure

TRANSACTION

Server: localhost » Database: nandinit » Table: TRANSACTION

BrowseStructureSQLSearchInsertExportImportOperationsTrackingTriggers

Showing rows 0 - 24 (119 total, Query took 0.0006 seconds.)

SELECT * FROM `TRANSACTION`

1 > >> Show all Number of rows: 25 Filter rows: Search this table

Sort by key: None

+ Options

		transaction_id	timestamp	total_amount	ticket_quantity	session_id
☐	Edit Copy Delete	T000001	2025-12-20 09:12:00	100.00	4	S10092
☐	Edit Copy Delete	T000002	2025-12-20 09:13:00	100.00	4	S11062
☐	Edit Copy Delete	T000003	2025-12-20 09:15:00	50.00	2	S10644
☐	Edit Copy Delete	T000004	2025-12-20 09:18:00	50.00	2	S10772
☐	Edit Copy Delete	T000005	2025-12-20 09:36:00	100.00	4	S11245
☐	Edit Copy Delete	T000006	2025-12-20 09:38:00	100.00	4	S10049
☐	Edit Copy Delete	T000007	2025-12-20 10:00:00	125.00	5	S10908
☐	Edit Copy Delete	T000008	2025-12-20 10:03:00	100.00	4	S10476
☐	Edit Copy Delete	T000009	2025-12-20 10:05:00	100.00	4	S10300
☐	Edit Copy Delete	T000010	2025-12-20 10:05:00	25.00	1	S10382
☐	Edit Copy Delete	T000011	2025-12-20 10:09:00	25.00	1	S10409
☐	Edit Copy Delete	T000012	2025-12-20 10:18:00	25.00	1	S10561
☐	Edit Copy Delete	T000013	2025-12-20 10:28:00	125.00	5	S11120
☐	Edit Copy Delete	T000014	2025-12-20 10:32:00	50.00	2	S10041
☐	Edit Copy Delete	T000015	2025-12-20 10:39:00	75.00	3	S11236
☐	Edit Copy Delete	T000016	2025-12-20 10:49:00	50.00	2	S10181
☐	Edit Copy Delete	T000017	2025-12-20 10:58:00	50.00	2	S10205
☐	Edit Copy Delete	T000018	2025-12-20 10:58:00	75.00	3	S10510
☐	Edit Copy Delete	T000019	2025-12-20 11:01:00	75.00	3	S10218
☐	Edit Copy Delete	T000020	2025-12-20 11:04:00	50.00	2	S10008
☐	Edit Copy Delete	T000021	2025-12-20 11:07:00	100.00	4	S10206
☐	Edit Copy Delete	T000022	2025-12-20 11:11:00	100.00	4	S10067
☐	Edit Copy Delete	T000023	2025-12-20 11:18:00	25.00	1	S10642
☐	Edit Copy Delete	T000024	2025-12-20 11:21:00	50.00	2	S10972
☐	Edit Copy Delete	T000025	2025-12-20 11:22:00	125.00	5	S10507

Server: localhost » Database: nandinit » Table: TRANSACTION

Browse

Structure

SQL

Search

Insert

Export

Import

Operations

Tracking

Triggers

Table structure

Relation view

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
☐	1	transaction_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary More
☐	2	timestamp	timestamp	on update CURRENT_TIMESTAMP	No	CURRENT_TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Change Drop Primary More
☐	3	total_amount	decimal(10,2)		No	None		Change Drop Primary More
☐	4	ticket_quantity	int(11)		No	None		Change Drop Primary More
☐	5	session_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary More

↑

☐ Check all

With selected:

Browse

Change

Drop

Primary

Unique

Index

Add to central columns

Remove from central columns

Print view

Propose table structure

?

Track table

Move columns

Improve table structure

SESSION

Server: localhost » Database: nandinit » Table: SESSION		
Browse	Structure	SQL
Search	Insert	Export
Import	Operations	Tracking
Triggers		
Showing rows 0 - 24 (1259 total, Query took 0.0003 seconds.)		
SELECT * FROM `SESSION`		
1	>	>>
Number of rows: 25		Filter rows: Search this table
Sort by key: None		
+ Options		
← T →	session_id	user_id
☐ Edit Copy Delete	S10158	U0001
☐ Edit Copy Delete	S10713	U0001
☐ Edit Copy Delete	S10571	U0002
☐ Edit Copy Delete	S10505	U0004
☐ Edit Copy Delete	S11013	U0004
☐ Edit Copy Delete	S10136	U0006
☐ Edit Copy Delete	S10359	U0007
☐ Edit Copy Delete	S10275	U0008
☐ Edit Copy Delete	S10729	U0008
☐ Edit Copy Delete	S11164	U0008
☐ Edit Copy Delete	S10162	U0009
☐ Edit Copy Delete	S10310	U0009
☐ Edit Copy Delete	S10948	U0009
☐ Edit Copy Delete	S11172	U0009
☐ Edit Copy Delete	S11191	U0010
☐ Edit Copy Delete	S10552	U0011
☐ Edit Copy Delete	S11003	U0011
☐ Edit Copy Delete	S10899	U0012
☐ Edit Copy Delete	S11202	U0013
☐ Edit Copy Delete	S10355	U0014
☐ Edit Copy Delete	S10577	U0014
☐ Edit Copy Delete	S11173	U0014
☐ Edit Copy Delete	S10209	U0015
☐ Edit Copy Delete	S10262	U0015
☐ Edit Copy Delete	S10340	U0015

Server: localhost » Database: nandinit » Table: SESSION

[Browse](#)
[Structure](#)
[SQL](#)
[Search](#)
[Insert](#)
[Export](#)
[Import](#)
[Operations](#)
[Tracking](#)
[Triggers](#)

[Table structure](#)
[Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
☐	1	session_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary Unique Index Spatial Fulltext Distinct values More
☐	2	user_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary Unique Index Spatial Fulltext Distinct values More

[Check all](#)
With selected:
[Browse](#)
[Change](#)
[Drop](#)
[Primary](#)
[Unique](#)
[Index](#)
[Add to central columns](#)
[Remove from central columns](#)

[Print view](#)
[Propose table structure](#)
[Track table](#)
[Move columns](#)
[Improve table structure](#)

PAGEVISIT

Server: localhost » Database: nandinit » Table: PAGEVISIT

[Browse](#)
[Structure](#)
[SQL](#)
[Search](#)
[Insert](#)
[Export](#)
[Import](#)
[Operations](#)
[Tracking](#)
[Triggers](#)

Showing rows 0 - 24 (12707 total, Query took 0.0004 seconds.)

[SELECT * FROM `PAGEVISIT`](#)

1
>
>>
Number of rows: 25
Filter rows: Search this table

Sort by key: None

+ Options

		pagevisit_id	timestamp	click_type	buy_ticket_clicked	checkout_step	form_error	usd_amount	session_id	page_id	referral_page_id
☐	Edit Copy Delete	PV000001	2025-12-20 10:24:00	page_view	0		0	0.00	S10001	P007	
☐	Edit Copy Delete	PV000002	2025-12-20 10:25:07	search	0		0	0.00	S10001	P001	P007
☐	Edit Copy Delete	PV000003	2025-12-20 10:25:56	click_cta	0		0	0.00	S10001	P012	P001
☐	Edit Copy Delete	PV000004	2025-12-20 10:27:07	page_view	0		0	0.00	S10001	P012	P012
☐	Edit Copy Delete	PV000005	2025-12-20 10:27:42	click_cta	0		0	0.00	S10001	P003	P012
☐	Edit Copy Delete	PV000006	2025-12-20 10:28:14	page_view	0		0	0.00	S10001	P001	P003
☐	Edit Copy Delete	PV000007	2025-12-20 10:28:37	search	0		0	0.00	S10001	P007	P001
☐	Edit Copy Delete	PV000008	2025-12-20 10:29:05	page_view	0		0	0.00	S10001	P003	P007
☐	Edit Copy Delete	PV000009	2025-12-20 10:29:28	search	0		0	0.00	S10001	P003	P003
☐	Edit Copy Delete	PV000010	2025-12-20 10:30:38	click_cta	0		0	0.00	S10001	P009	P003
☐	Edit Copy Delete	PV000011	2025-12-20 10:32:04	search	0		0	0.00	S10001	P006	P009
☐	Edit Copy Delete	PV000012	2025-12-20 10:32:38	page_view	0		0	0.00	S10001	P007	P006
☐	Edit Copy Delete	PV000013	2025-12-20 10:33:24	page_view	0		0	0.00	S10001	P012	P007
☐	Edit Copy Delete	PV000014	2025-12-20 11:19:00	search	0		0	0.00	S10002	P012	
☐	Edit Copy Delete	PV000015	2025-12-20 11:20:18	click_cta	0		0	0.00	S10002	P002	P012
☐	Edit Copy Delete	PV000016	2025-12-20 11:21:00	search	0		0	0.00	S10002	P008	P002
☐	Edit Copy Delete	PV000017	2025-12-20 11:22:17	click_cta	0		0	0.00	S10002	P007	P008
☐	Edit Copy Delete	PV000018	2025-12-20 11:23:28	search	0		0	0.00	S10002	P012	P007
☐	Edit Copy Delete	PV000019	2025-12-20 11:24:43	page_view	0		0	0.00	S10002	P005	P012
☐	Edit Copy Delete	PV000020	2025-12-20 11:25:45	page_view	0		0	0.00	S10002	P005	P005
☐	Edit Copy Delete	PV000021	2025-12-20 11:26:49	search	0		0	0.00	S10002	P003	P005
☐	Edit Copy Delete	PV000022	2025-12-20 11:28:16	click_cta	0		0	0.00	S10002	P005	P003
☐	Edit Copy Delete	PV000023	2025-12-20 11:28:56	click_cta	0		0	0.00	S10002	P002	P005
☐	Edit Copy Delete	PV000024	2025-12-20 11:29:53	page_view	0		0	0.00	S10002	P008	P002
☐	Edit Copy Delete	PV000025	2025-12-20 11:30:31	search	0		0	0.00	S10002	P002	P008

Server: localhost » Database: nandinit » Table: PAGEVISIT

[Browse](#)
[Structure](#)
[SQL](#)
[Search](#)
[Insert](#)
[Export](#)
[Import](#)
[Operations](#)
[Tracking](#)
[Triggers](#)

[Table structure](#)
[Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
☐	1	pagevisit_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary More
☐	2	timestamp	timestamp	on update CURRENT_TIMESTAMP	No	CURRENT_TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Change Drop Primary More
☐	3	click_type	varchar(20)	utf8mb4_bin	No	None		Change Drop Primary More
☐	4	buy_ticket_clicked	tinyint(1)		No	None		Change Drop Primary More
☐	5	checkout_step	varchar(20)	utf8mb4_bin	No	None		Change Drop Primary More
☐	6	form_error	tinyint(1)		No	None		Change Drop Primary More
☐	7	usd_amount	decimal(10,2)		No	None		Change Drop Primary More
☐	8	session_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary More
☐	9	page_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary More
☐	10	referral_page_id	varchar(10)	utf8mb4_bin	No	None		Change Drop Primary More

☐ Check all
 With selected:
 [Browse](#)
[Change](#)
[Drop](#)
[Primary](#)
[Unique](#)
[Index](#)
[Add to central columns](#)
[Remove from central columns](#)

[Print view](#)
[Propose table structure](#)
[Track table](#)
[Move columns](#)
[Improve table structure](#)

PAGE

Server: localhost » Database: nandinit » Table: PAGE

[Browse](#)
[Structure](#)
[SQL](#)
[Search](#)
[Insert](#)
[Export](#)
[Import](#)
[Operations](#)
[Tracking](#)
[Triggers](#)

Showing rows 0 - 11 (12 total, Query took 0.0005 seconds.)

SELECT * FROM `PAGE`

☐ Show all
 Number of rows: 25
 Filter rows: Search this table

Sort by key: None

+ Options

			page_id	name	description	
☐	Edit	Copy	Delete	P001	donate	The Donate page provides a simple and secure way f...
☐	Edit	Copy	Delete	P002	education	The Education page highlights learning programs, w...
☐	Edit	Copy	Delete	P003	exhibition_detail	The Exhibition Detail page provides in-depth infor...
☐	Edit	Copy	Delete	P004	exhibition_list	The Exhibition List page showcases current, and up...
☐	Edit	Copy	Delete	P005	generic_content	The Generic Content page presents flexible informa...
☐	Edit	Copy	Delete	P006	home	The Home page introduces visitors to the museum, f...
☐	Edit	Copy	Delete	P007	map_directions	The Map & Directions page offers location details,...
☐	Edit	Copy	Delete	P008	membership	The Membership page outlines membership benefits, ...
☐	Edit	Copy	Delete	P009	search_results	The Search Results page displays relevant content ...
☐	Edit	Copy	Delete	P010	tickets_checkout	The Tickets Checkout page guides visitors through ...
☐	Edit	Copy	Delete	P011	tickets_overview	The Tickets Overview page provides information on ...
☐	Edit	Copy	Delete	P012	visit_info	The Visit Info page shares essential details for p...

Server: localhost » Database: nandinit » Table: PAGE

[Browse](#)
[Structure](#)
[SQL](#)
[Search](#)
[Insert](#)
[Export](#)
[Import](#)
[Operations](#)
[Tracking](#)
[Triggers](#)

[Table structure](#)
[Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
☐ 1	page_id	varchar(10)	utf8mb4_bin		No	None		Change Drop Primary Unique Index Spatial Fulltext Distinct values More
☐ 2	name	varchar(50)	utf8mb4_bin		No	None		Change Drop Primary Unique Index Spatial Fulltext Distinct values More
☐ 3	description	varchar(255)	utf8mb4_bin		No	None		Change Drop Primary Unique Index Spatial Fulltext Distinct values More

☐ Check all
 With selected:
 [Browse](#)
[Change](#)
[Drop](#)
[Primary](#)
[Unique](#)
[Index](#)
[Add to central columns](#)
[Remove from central columns](#)

[Print view](#)
[Propose table structure](#)
[Track table](#)
[Move columns](#)
[Improve table structure](#)

SQL Analysis

Funnel Diagnosis

```
SELECT COUNT(session_id) AS 'Total Sessions Analyzed', AVG(step_count) AS 'Avg Steps (All Sessions)', AVG(CASE WHEN reached_tickets = 1 THEN step_count END) AS 'Avg Steps (Ticket Sessions)', AVG(CASE WHEN converted = 1 THEN step_count END) AS 'Avg Steps (Converted Sessions)' FROM ( SELECT pv.session_id, COUNT(*) AS step_count, MAX(CASE WHEN p.name IN ('tickets_overview','tickets_checkout') THEN 1 ELSE 0 END) AS reached_tickets, MAX(CASE WHEN pv.checkout_step = 'complete' THEN 1 ELSE 0 END) AS converted FROM PAGEVISIT pv INNER JOIN PAGE p ON pv.page_id = p.page_id GROUP BY pv.session_id ) AS session_metrics
```

+ Options

Total Sessions Analyzed	Avg Steps (All Sessions)	Avg Steps (Ticket Sessions)	Avg Steps (Converted Sessions)
1259	10.0929	10.1747	10.0672

Analysis

1. Does the museum's funnel exceed the industry benchmark of 5.1 steps?

Yes The average funnel length across all sessions is 10.09 steps, almost double the industry benchmark of 5.1 checkout steps.

2. Two possible reasons the funnel exceeds 5.1 steps:

- Users browse before purchasing. The data shows that only 372 out of 1,259 sessions (29.5%) ever reached the tickets_overview page. The remaining users spend many steps on content pages before going toward the ticket purchase.
- Repeated checkout attempts due to form errors. All 61 form error events occurred on the tickets_checkout page.

3. Comparison to the "how tall is the funnel" result from Part 3:

The Part 3 query returned an average of 10.09 steps across all sessions. The funnel diagnosis query above returns the same value (~10.09) for all sessions, and 10.17 specifically for ticket-related sessions. These are almost the same.

4. Why might the two averages differ slightly?

The Part 3 query averages step counts across all 1,259 sessions regardless of behavior. The funnel diagnosis query segments sessions by whether they reached a ticket page or converted.

Advanced Funnel Diagnosis (Errors + User Type)

```
SELECT
u.user_type,
COUNT(DISTINCT s.session_id) AS 'Total Sessions',
SUM(sm.converted) AS 'Conversions',
ROUND(SUM(sm.converted) / COUNT(DISTINCT s.session_id) * 100, 2)
AS 'Conversion Rate (%)',
ROUND(AVG(sm.step_count), 2) AS 'Avg Steps',
ROUND(AVG(CASE WHEN sm.had_error = 1
THEN sm.step_count END), 2) AS 'Avg Steps (With Error)',
ROUND(AVG(CASE WHEN sm.had_error = 0
THEN sm.step_count END), 2) AS 'Avg Steps (No Error)',
SUM(sm.had_error) AS 'Sessions With Errors',
MAX(sm.step_count) AS 'Max Steps in Any Session'
FROM USER u
INNER JOIN SESSION s ON u.user_id = s.user_id
INNER JOIN (
SELECT
session_id,
COUNT(*) [...]
```

Options

user_type	Total Sessions	Conversions	Conversion Rate (%)	Avg Steps	Avg Steps (With Error)	Avg Steps (No Error)	Sessions With Errors	Max Steps in Any Session
guest	960	94	9.79	10.06	9.75	10.08	40	14
member	144	11	7.64	10.28	10.67	10.26	6	14
student	155	14	9.03	10.12	11.33	10.07	6	14

Analysis

1. Do user types show meaningful differences in conversion rate or average funnel length?

No, the two user types are very close. Guest users convert at 9.44% and registered users at 9.49%. Average funnel length is also close: 10.08 steps for guests vs. 10.12 for registered users.

2. Which user type experiences more drop-off or errors, and at what stage?

Guest users account for more sessions with errors (40 vs. 12), which is proportional to their share of overall sessions (76.5% of all sessions). All form errors happen on the tickets_checkout page. The more bigger drop happens before the checkout page: of 1,259 total sessions, only 372 (29.5%) ever reach the tickets_overview page. Of those, only 119 (32.0%) proceed to complete a purchase.

Funnel Exploratory Analysis

QUESTION: Which entry page leads to the highest ticket conversion rate?

```

SELECT
entry_page.name AS 'Entry Page',
COUNT(DISTINCT sm.session_id) AS 'Sessions Starting Here',
SUM(sm.converted) AS 'Conversions',
ROUND(SUM(sm.converted) /
COUNT(DISTINCT sm.session_id) * 100, 2) AS 'Conversion Rate (%)',
ROUND(AVG(sm.step_count), 2) AS 'Avg Steps'
FROM (
SELECT pv.session_id, pv.page_id
FROM PAGEVISIT pv
INNER JOIN (
SELECT session_id, MIN(timestamp) AS first_time
FROM PAGEVISIT
GROUP BY session_id
) AS first_ts
ON pv.session_id = first_ts.session_id
AND pv.timestamp = first_ts.first_time
) AS first_visit
INNER JOIN PAGE entry_page ON first_visit.page_id = entry_page.page_id
INNER JOIN (
SELECT
session_id,
COUNT(*) AS step_count,
MAX(CASE WHEN checkout_step = 'comple[...]
```

+ Options

Entry Page	Sessions Starting Here	Conversions	Conversion Rate (%) ▾ 1	Avg Steps
exhibition_list	113	16	14.16	9.69
membership	119	15	12.61	9.91
map_directions	126	15	11.90	9.97
education	128	13	10.16	10.27
exhibition_detail	138	14	10.14	9.83
home	115	9	7.83	10.10
generic_content	131	10	7.63	10.02
visit_info	131	10	7.63	10.37
donate	132	9	6.82	10.45
search_results	126	8	6.35	10.27

Analysis

The entry page a user lands on has a big effect on whether they buy a ticket. Users entering through `exhibition_list` converted at the highest rate (14.2%), followed by membership (12.6%) and `map_directions` (11.9%). Users entering through search results (6.3%), donate (6.8%), and generic content pages (7.6%) converted at roughly half the rate.

Exhibition list and membership pages attract users with more intent since they are actively exploring what the museum offers.. The home page, despite being the most common expected entry point for direct visitors, converts at only 7.8%. The recommendation is to put "Buy Tickets" CTAs on `exhibition_list` and membership pages and redesign the homepage and `search_results` to have a more convincing path to purchasing a ticket.

Written Analysis and Findings

ANALYSIS 1: FUNNEL LENGTH VS INDUSTRY BENCHMARK

What We Measured

The first query: Is the museum's ticket purchase funnel longer than the industry standard, and does this vary depending on whether a user reached the ticket pages or completed a purchase?

The query calculated the average number of page visit events (steps) per session across three groups: all sessions, sessions that reached either the tickets_overview or tickets_checkout page, and sessions that ended in a completed purchase.

Key Results

Metric	Value
Avg steps — all sessions	10.09
Avg steps — ticket-page sessions	10.17
Avg steps — converted sessions	10.07
Industry benchmark (ConvertCart)	5.1 steps
Funnel excess vs. benchmark	+97% longer
Sessions reaching ticket pages	372 / 1,259 (29.5%)
Overall conversion rate	119 / 1,259 = 9.45%

Interpretation and Conclusion

The museum's funnel is almost twice as long as the 5.1-step industry benchmark (10.09 steps across all sessions). So this is not a matter of users wandering around the site and then leaving. Converted sessions average 10.07 steps, almost identical to the overall average. This means that even users who successfully purchased a ticket took twice as many steps as they should have had to.

Only 372 out of 1,259 sessions (29.5%) ever reached the tickets_overview or tickets_checkout page. The remaining 70.5% of sessions never entered the purchase funnel. Users are spending their steps browsing content pages (exhibition_detail, education, membership, map_directions) without being guided toward ticket purchase. This suggests the 'Buy Tickets' prompt is not prominent enough throughout the site. A user who has to navigate through seven or eight content pages before discovering the ticket option would probably leave before converting. The redesign

should prioritize placing ticket purchase prompts on every major content page, not just the homepage.

ANALYSIS 2: CHECKOUT ERROR AND USER TYPE

What We Measured

The second query asked: Do checkout form errors increase funnel length and reduce conversion rates, and does this effect differ between guest and registered users? The query joined the USER, SESSION, and PAGEVISIT tables to compute, for each user type, the total number of sessions, conversions, conversion rate, average steps, average steps with and without errors, and the number of sessions that encountered at least one form error.

Key Results

Metric	Value
Guest sessions	964
Guest conversion rate	9.44%
Guest avg steps	10.08
Guest sessions with form errors	40 (4.1%)
Registered sessions	295
Registered conversion rate	9.49%
Registered avg steps	10.12
Registered sessions with form errors	12 (4.1%)
Total form error events	61 — all on tickets_checkout
Sessions affected by errors	52

Interpretation and Conclusion

The biggest finding from this analysis is how little difference exists between guest and registered users. Their conversion rates are almost identical (9.44% vs. 9.49%), and their average funnel lengths differ by only 0.04 steps. Usually, registered users convert at higher rates because they have saved payment information and a relationship with the brand. The absence of this gap means that the checkout experience is not different enough between the two groups.

Form errors occur at an identical rate for both user types (4.1% of sessions) and are on the tickets_checkout page, but errors are not the primary reason for poor conversion. The conversion

rate for sessions with errors is actually non-zero. 70.5% of sessions never reach the ticket funnel at all. The redesign should address checkout error messaging.

ANALYSIS 3: CONVERSION RATE BY ENTRY PAGE

What We Measured

The third query asked: Which page a user lands on first has the greatest impact on whether they purchase a ticket? The query identified each session's first page visit using the earliest timestamp, then produced a conversion rate and average steps broken down by entry page.

Key Results

Metric	Value
exhibition_list — conversion rate	14.2% (16 / 113 sessions)
membership — conversion rate	12.6% (15 / 119 sessions)
map_directions — conversion rate	11.9% (15 / 126 sessions)
home — conversion rate	7.8% (9 / 115 sessions)
donate — conversion rate	6.8% (9 / 132 sessions)
search_results — conversion rate	6.3% (8 / 126 sessions)
Spread (highest vs. lowest entry)	14.2% vs. 6.3% — 2.3× difference
Avg steps across all entry pages	~10.0–10.2 (uniform)

Interpretation and Conclusion

Entry page has a big effect on conversion. Users landing on the exhibition_list page convert at 14.2%, which is more than double the rate of users entering through search_results (6.3%) or donate (6.8%). The homepage converts at only 7.8%, below the dataset average of 9.45%. This is a usability red flag.

The pattern across high-converting entry pages is important. exhibition_list, membership, and map_directions all reflect users with the intention to buy.

By contrast, users entering through search_results or donate are less committed. The homepage's underperformance is the biggest finding. The site's home is not doing enough to move visitors toward the purchasing funnel. Average session depth is almost identical across all entry pages (9.97 to 10.21 steps), so these conversion differences are driven by where users start, not by how

long they browse. The redesign should treat pages with a high conversion rate (like exhibition list and membership) as primary conversion places, with ticket purchase prompts put into their layouts.